DevOpsCon

by entwickler.de

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Building an Event-Driven CI/CD Provenance System

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Event Driven CI/CD Workshop

View the workshop:

https://github.com/xbcsmith/epr-workshop

Clone the workshop:

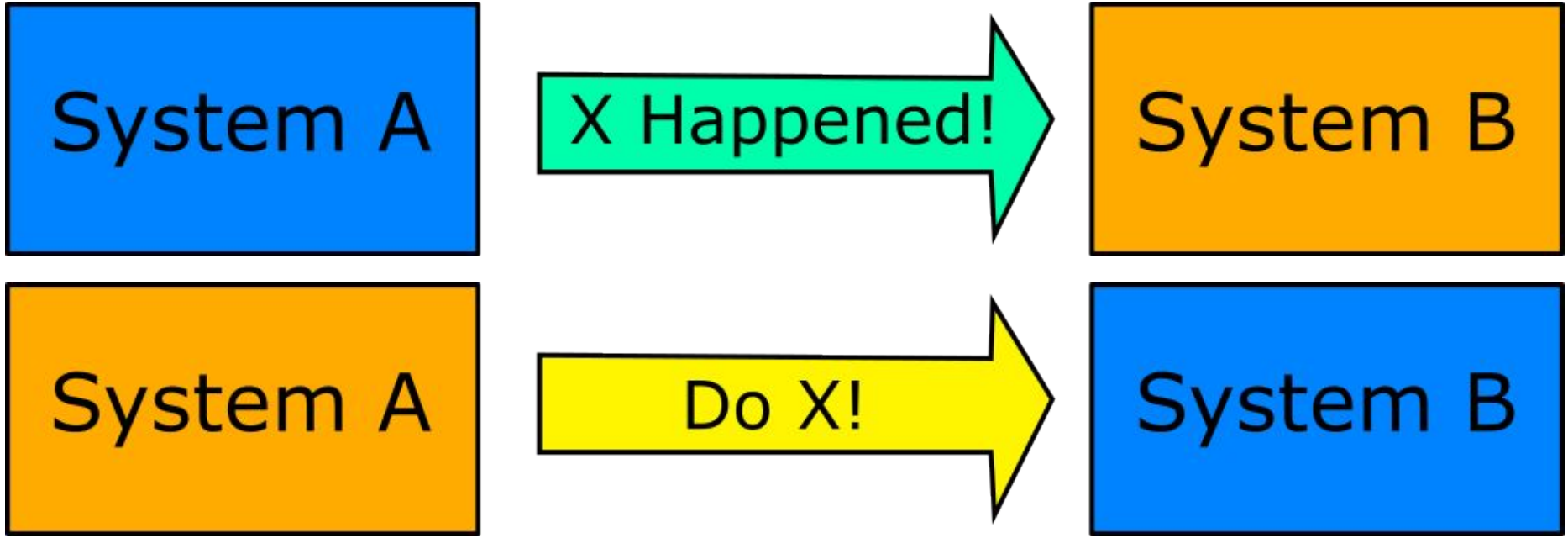
```
git clone git@github.com:xbcsmith/epr-workshop.git
```

Introduction to Event Driven Microservice Architectures

- Asynchronous
- Allows for scalability and modularity
- Provides an audit trail
- Services are specialized
- Independent SDLC

- Creeping system complexity
- Eventual consistency
- Increased maintenance
- Downstream flooding
- Dead Letters

Declarative and Imperative Events



Cloud Events

- Interoperability
- Producer Consumer Independence
- Asynchronous Communication
- Standard Data Model

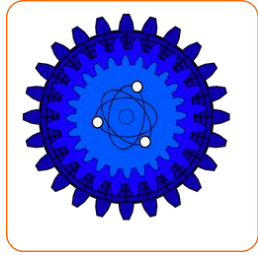
- Dynamic Event Interest Expression
- Flexible Event Consumption
- Consumer Evolution

CDEvents

- Common Vocabulary
- Normalized Form
- System Agnostic

- Declarative events
- Simplified Integration
- Targeted Notifications

Event Provenance Registry (EPR)



- Events
- Event Receivers
- Event Receiver Groups

- Gating
- Auditing
- Provenance

- Messages
- CloudEvents Spec
- Declarative events
- Imperative events
- Watchers

EPR
Workshop

Setting Up the Environment

Docker – https://docs.docker.com/get-docker

Docker Compose – https://docs.docker.com/compose/install

Golang – https://go.dev/doc/install

Python – https://www.python.org/downloads

Git – https://git-scm.com/downloads

Redpanda (Kafka): Message Queue Overview

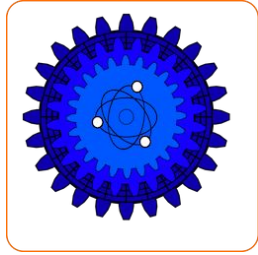
Redpanda is a Kafka-compatible event streaming platform.

It provides a message bus for our event-driven architecture.

Simple. Kafka® API-compatible. ZooKeeper® free. JVM free.

- Redpanda Nodes - Self contained service
- Redpanda Keeper - CLI for managing clusters
- Redpanda Console - UI for managing clusters

Redpanda (Kafka): Message Queue Overview



Terminal 1 : run the command to consume the message

```
docker compose -f ./compose/docker-compose.yaml up
```

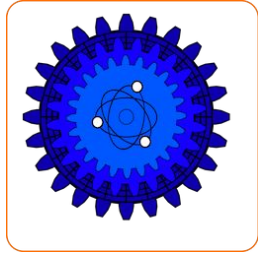
Terminal 2 : run the commands create a topic and produce the message

```
docker exec -it redpanda rpk topic create epr.dev.events --brokers=localhost:9092
docker exec -it redpanda rpk topic produce epr.dev.events --brokers=localhost:9092
```

Terminal 3 : run the command to consume the message

```
docker exec -it redpanda rpk topic consume epr.dev.events --brokers=localhost:9092
```

EPR Concepts



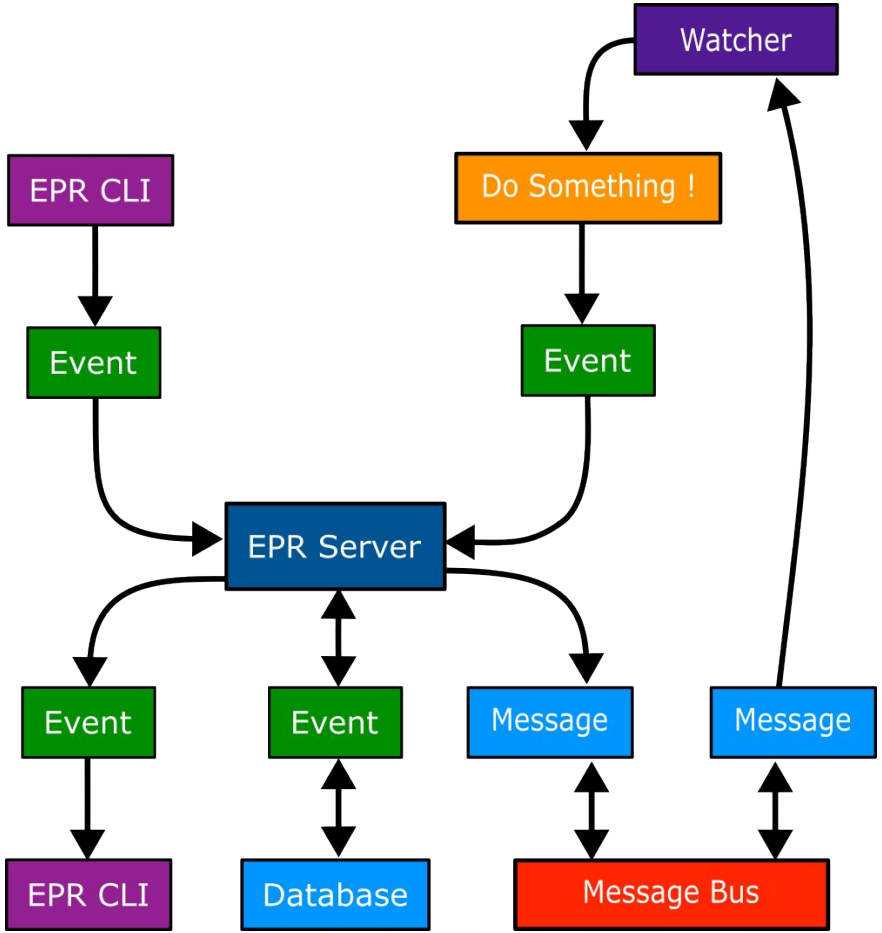
- Events
- Event Receivers
- Event Receiver Groups

- Gating
- Auditing
- Provenance

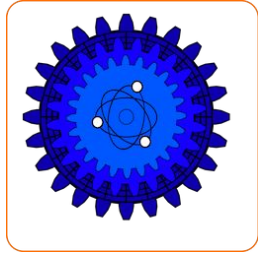
- N - Name
- V - Version
- R - Release
- P - Platform ID
- P - Package

- Messages
- CloudEvents Spec
- Declarative events
- Imperative events
- Watchers

EPR Workflow



Exploring EPR: Codebase Overview



Use the following command to clone the EPR project repository:

```
git clone git@github.com:sassoftware/event-provenance-registry.git
```

Exploring EPR: Start the Backend

Open a new terminal and cd to the event-provenance-registry repository.

```
cd event-provenance-registry
```

Use the following command to start the backend services for EPR:

```
docker compose -f ./docker-compose.services.yaml up
```

Building EPR: Service

Open a new terminal and cd to the event-provenance-registry repository.

```
cd event-provenance-registry
```

Export the following environment variables:

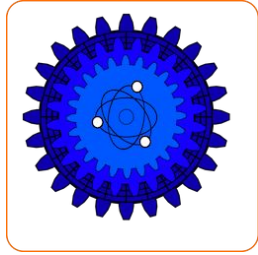
```
export EPR_TOPIC=epr.dev.events
export EPR_BROKERS=localhost:19092
export EPR_DB=postgres://localhost:5432
```

Then we can start the EPR server with the following command:

```
go run main.go
```

The server will be available on `localhost:8042`

Setup EPR: Workflow



EPR workflow to setup the server to produce events is as follows:

1. Create an Event Receiver.
2. Post an Event to the Event Receiver.
3. EPR produces message on topic.

To leverage Event Receiver Groups we would follow this workflow:

1. Create several Event Receivers
2. Create an Event Receiver Group with all the Event Receivers.
3. Create an Event with identical NVRPP for each Event Receiver in the Group.
4. EPR produces a message for each Event on the topic.
5. When the last Event is sent EPR produces a message for the Event Receiver Group on the topic.

Curl EPR: Create an Event Receiver

```
curl --location --request POST 'http://localhost:8042/api/v1/receivers' \
--header 'Content-Type: application/json' \
--data-raw '{
 "name": "foobar",
 "type": "foo.bar",
 "version": "1.1.3",
 "description": "The event receiver of Brixton",
 "schema": {
   "type": "object",
   "properties": {
     "name": {
       "type": "string"
     }
   }
 }
}'
```

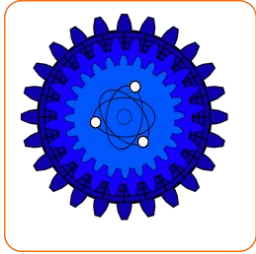
Curl EPR: Create an Event

```
curl --location --request POST 'http://localhost:8042/api/v1/events' \
--header 'Content-Type: application/json' \
--data-raw '{
 "name": "magnificent",
 "version": "7.0.1",
 "release": "2023.11.16",
 "platform_id": "linux",
 "package": "docker",
 "description": "blah",
 "payload": {
   "name": "joe"
 },
 "success": true,
 "event_receiver_id": "<PASTE EVENT RECEIVER ID FROM FIRST CURL COMMAND>"
}
```

Curl EPR: Create an Event Receiver Group

```
curl --location --request POST 'http://localhost:8042/api/v1/groups'
\
--header 'Content-Type: application/json'  \
--data-raw '{
 "name": "the_clash",
 "type": "foo.bar",
 "version": "3.3.3",
 "description": "The only event receiver group that matters",
 "enabled": true,
 "event_receiver_ids": [
   "PASTE EVENT RECEIVER ID FROM FIRST CURL COMMAND"
 ]
}
```

Using EPR: GraphQL

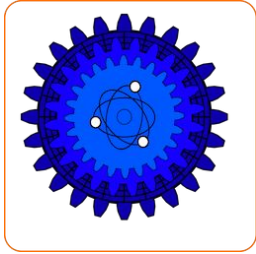


The GraphQL Playground is a tool that allows you to test your GraphQL queries in a browser.

The graphql playground will now be accessible at:

```
http://localhost:8042/api/v1/graphql
```

Using EPR: GraphQL



Create an Event Receiver by pasting the mutation into the GraphQL Playground.

```
mutation {
 create_event_receiver(
   event_receiver: {
     name: "the_clash"
     version: "1.0.0"
     type: "london.calling"
     description: "The only band that matters"
     schema: "{\"name\": \"value\"}"
   }
 )
}
```

Using EPR: GraphQL

Create an Event Receiver Group by pasting the mutation into the GraphQL Playground. Use the ID from the last command for the event_receivers_ids field.

```
mutation {
 create_event_receiver_group(
   event_receiver_group: {
     name: "foobar"
     version: "1.0.0"
     description: "a fake event receiver group"
     enabled: true
     event_receiver_ids: ["ID_RETURNED_FROM_PREVIOUS_MUTATION"]
     type: "test.test.test"
   }
 )
}
```

Using EPR: GraphQL

Create an Event by pasting the mutation into the GraphQL Playground.

```
mutation {
 create_event(
   event: {
     name: "foo"
     version: "1.0.0"
     release: "20231103"
     platform_id: "platformID"
     package: "package"
     description: "The Foo of Brixton"
     payload: "{\"name\": \"value\"}"
     event_receiver_id: "ID_RETURNED_FROM_PREVIOUS_MUTATION"
     success: true
   }
 )
}
```

Building EPR: CLI

Open a new terminal and cd to the cli directory in the event-provenance-registry repository.

```
cd event-provenance-registry/cli
```

Build CLI with the following commands:

Linux

```
make PREFIX=$(go env GOPATH) install
```

Mac OS X M1

```
make PREFIX=$(go env GOPATH) install-darwin-arm64
```

Export the following environment variables:

```
export EPR_TOPIC=epr.dev.events
export EPR_BROKERS=localhost:19092
export EPR_DB=postgres://localhost:5432
```

Using EPR: CLI

Export the following environment variables:

```
export EPR_TOPIC=epr.dev.events
```

```
export EPR_BROKERS=localhost:19092
```

```
export EPR_DB=postgres://localhost:5432
```

Using EPR: CLI

Create a pair of Event Receivers

```
epr-cli receiver create \
     --name "foo-cli" \
     --version "1.0.0" \
     --description "foo cli created foo" \
     --type "epr.foo.cli" \
     --schema "{}"
```

```
epr-cli receiver create \
     --name "bar-cli" \
     --version "1.0.0" \
     --description "bar cli created bar" \
     --type "epr.bar.cli" \
     --schema "{}"
```

Using EPR: CLI

Create an Event Receiver Group. Use the IDs from the Event Receivers

```
epr-cli group create \
     --name "foo-group-cli" \
     --version "1.0.0" \
     --description "foo cli created foo group" \
     --type "epr.foo.group.cli" \
     --event-receiver-ids "01HKX0J9KS8AASMRYX61458N41 01HKX0KY3B31MR3XKJWTDZ4EQ0"
```

Using EPR: CLI

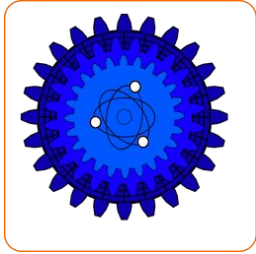
Search for objects by ID, or Name and Version, or Type.

```
epr-cli receiver search --id 01HKX0KY3B31MR3XKJWTDZ4EQ0 --fields all

epr-cli receiver search --type epr.foo.cli --fields all

epr-cli receiver search --name foo-cli --version 1.0.0 --fields all
```

Extended EPR: Creating a watcher



Use the EPR watcher SDK to craft a watcher, which actively listens for events originating from the EPR server.

Make a new directory for your watcher and create a `main.go` in that directory.

```
mkdir foo
cd foo
touch main.go
```

Extended EPR: Creating a watcher

Now open the `main.go` in your favorite editor (Vim).

Add the following code:

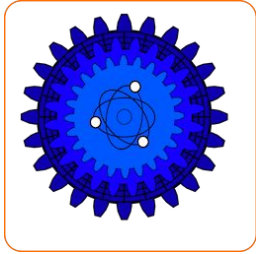
```go
package main

import (
"encoding/json"
"log"

"github.com/sassoftware/event-provenance-registry/pkg/message"
"github.com/sassoftware/event-provenance-registry/pkg/watcher"
)

func main() {
seeds := []string{"localhost:19092"}
topics := []string{"epr.dev.events"}
consumerGroup := "watcher-workshop"
watcher, err := watcher.New(seeds, topics, consumerGroup)
if err != nil {
 panic(err)
}
defer watcher.Client.Close()
go watcher.StartTaskHandler(customTaskHandler)
watcher.ConsumeRecords(customMatcher)
}
```

Extended EPR: Creating a watcher



Add:

- customMatcher
  - matching the message type value.
- customTaskHandler
  - prints a message when a match is found.

```go
func customMatcher(msg *message.Message) bool {
return msg.Type == "foo.bar"
}

func customTaskHandler(msg *message.Message) error {
log.Default().Printf("I received a task with value ' %v'", msg)
return nil
}
```

Save the file and run `go mod init` in your terminal.

Run `go mod tidy` to fill in the dependencies.

Extended EPR: Creating a watcher

Start up the watcher and start consuming messages with the following command:

```
go run main.go
```

In a second terminal create an Event Receiver where type matches what you put in the customMatcher.

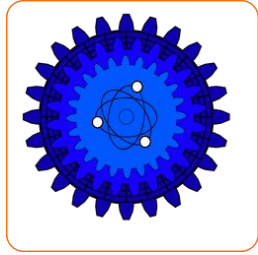
```
curl --location --request POST 'http://localhost:8042/api/v1/receivers'\
--header 'Content-Type: application/json'\
--data-raw '{
 "name": "watcher-workshop",
 "type": "foo.bar",
 "version": "1.0.0",
 "description": "The event receiver of Brixton",
 "enabled": true,
 "schema": {
 "type": "object",
 "properties": {
   "name": {
     "type": "string"
   }
 }
}
}'
```

Extended EPR: Creating a watcher

Now create an Event.

```
curl --location --request POST 'http://localhost:8042/api/v1/events'  \
--header 'Content-Type: application/json'  \
--data-raw '{
   "name": "magnificent",
   "version": "7.0.1",
   "release": "2023.11.16",
   "platform_id": "linux",
   "package": "docker",
   "description": "blah",
   "payload": {"name":"joe"},
   "success": true,
   "event_receiver_id": "<PASTE EVENT RECEIVER ID FROM FIRST CURL COMMAND>"
}'
```

Advanced EPR: CDEvents

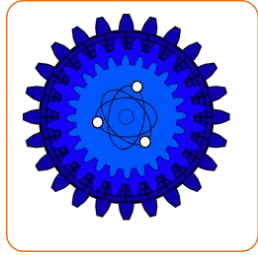


In this section we will cover the CDEvents and EPR concepts and how to use them.

- Event Receiver Schema
- Create an Event Receiver with a CDEvent schema
- Post an Event without a payload and a CDEvent payload
- Create a watcher to trigger on CDEvents types

The commands in this section are long. Please refer to section 07-epr-cdevents

Advanced EPR: SBOMs

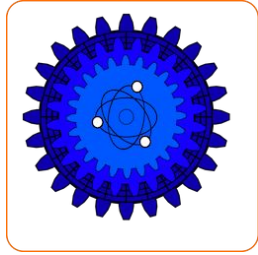


In this section of the workshop, we will introduce the SBOMs. We will follow a pattern similar to the previous CDEvents section.

- Event Receiver Schema
- CycloneDX bom schema
- Create a source SBOM event
- Create an SBOM for an OCI image

The commands in this section are long. Please refer to section 08-epr-sboms

Advanced EPR: Python

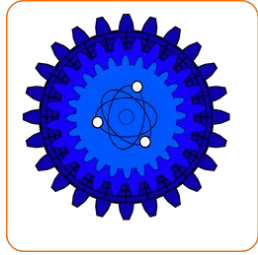


In this section we will walk through the development and usage of the EPR Python client.

- EPR Python Client
- Overview
- Development
- Model Example
- Client Create Example
- Create an event receiver
- Create an event receiver group
- Create an event
- EPR Python CLI

The commands in this section are long. Please refer to section 09-epr-python

Advanced EPR: Workshop Challenge

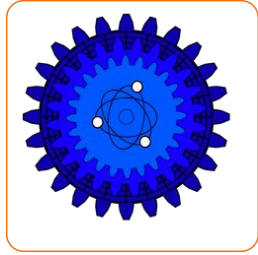


In this section we will expand on our EPR concepts we have learned in the previous sections. This is a free form exercise to expand on what we have learned.

Suggestions:

- Add a watcher to fire a webhook.
- Add a watcher to create a CDEvent.
- Add a watcher to create a SBOM.
- Add a Python watcher to create an Event.

AMA



I am Smitty and I am afraid of Robots.

EPR Workshop

EPR

EPR-Python

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Please,

don't forget

to vote!

devmio